

Five places, only four tests

Name: _____

Date: _____

Write R as 0 and B as 1. A score still counts correct **positions**. Choose all tests in advance: 00000, 00011, 00101, 01001.

1. Decode a real example.

These four tests score 2, 2, 2, 0 in that order. Find the secret. Verify every score.

Secret and check:

2. Why could this always work?

Let the secret be $x_1x_2x_3x_4x_5$, with each x_i equal to 0 or 1. The first score tells the total number of 1s. What does changing two test positions from 0 to 1 tell you?

Relate the score changes to these sums: $x_4 + x_5$, $x_3 + x_5$, $x_2 + x_5$.

3. The only tricky case.

A pair-sum of 0 or 2 forces both bits. What if all three pair-sums are 1? Try $x_5 = 0$ and $x_5 = 1$. Can they have the same total number of 1s, even after you choose x_1 ?

Why the four answers identify every possible secret:

A geometric translation: Join binary words that differ in one position. Distance is the number of mismatches, so distance = 5 minus score. These four test words are landmarks that distinguish all 32 vertices of a five-dimensional cube.

Build first. Explain by pointing, talking, or drawing.